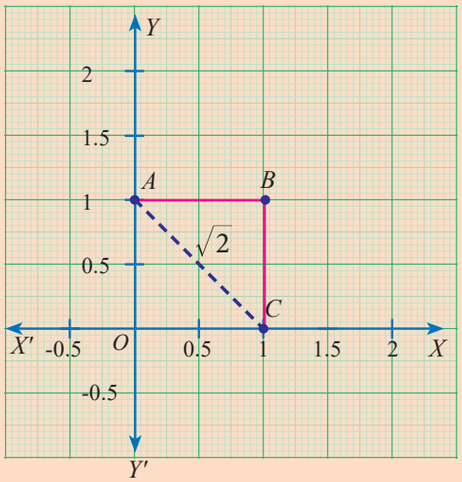
Activity - 2

Take a graph sheet and mark O, A, B, C as follows.



In the square $OABC$,

$OA = AB = BC = OC = 1$unit

Consider right angled ΔOAC

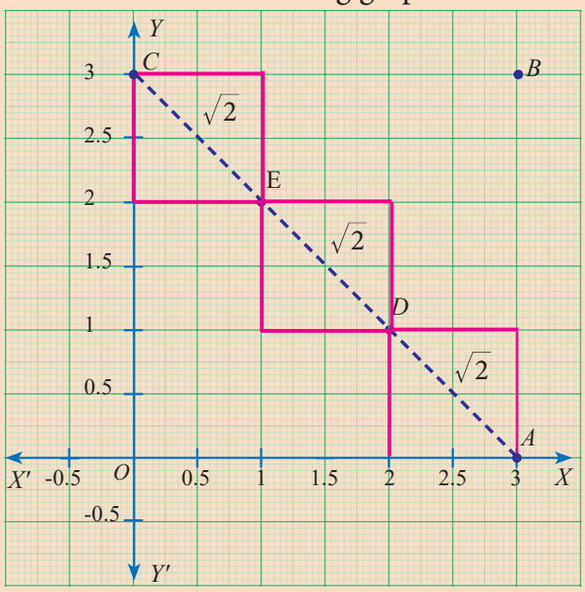
$$AC = \sqrt{1^2 + 1^2}$$

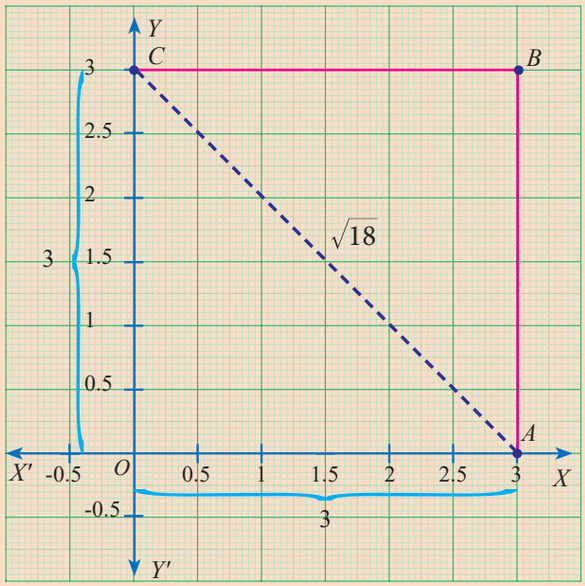
$$= \sqrt{2} \text{ unit [By Pythagoras theorem]}$$

The length of the diagonal (hypotenuse)

$AC = \sqrt{2}$, which is a surd.

Consider the following graphs:





Let us try to find the length of AC in two different ways :

$AC = AD + DE + EC$

(diagonals of units squares)

$= \sqrt{2} + \sqrt{2} + \sqrt{2}$

$AC = 3\sqrt{2}$ units

$AC = \sqrt{OA^2 + OC^2} = \sqrt{3^2 + 3^2}$

$= \sqrt{9 + 9}$

$AC = \sqrt{18}$ units

Are they equal? Discuss. Can you verify the same by taking different squares of different lengths?

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